yEvo Mutation Browser

Overview

The yEvo Mutation Browser is an R Shiny app that enables users to visualize mutations in the *Saccharomyces cerevisiae* (*S. cerevisiae*) genome via several interactive graphs, including a chromosome map, mutation spectra, pie chart of mutation types, and gene view. This was built for high school students in the yEvo program, who have experimentally evolved yeast to adapt to various conditions. Therefore, all cumulative yEvo data is bundled within the app and is always available there. However, users outside of yEvo can also use this tool to upload their own mutation data and use the interactive visualizations, making it a tool for both educational and research use.

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Usage

The easiest method to use the mutation browser is to access it via the web at https://yevo.org/mutation-browser/ or via Shiny apps. To run the mutation browser locally, see the section Running Locally below.

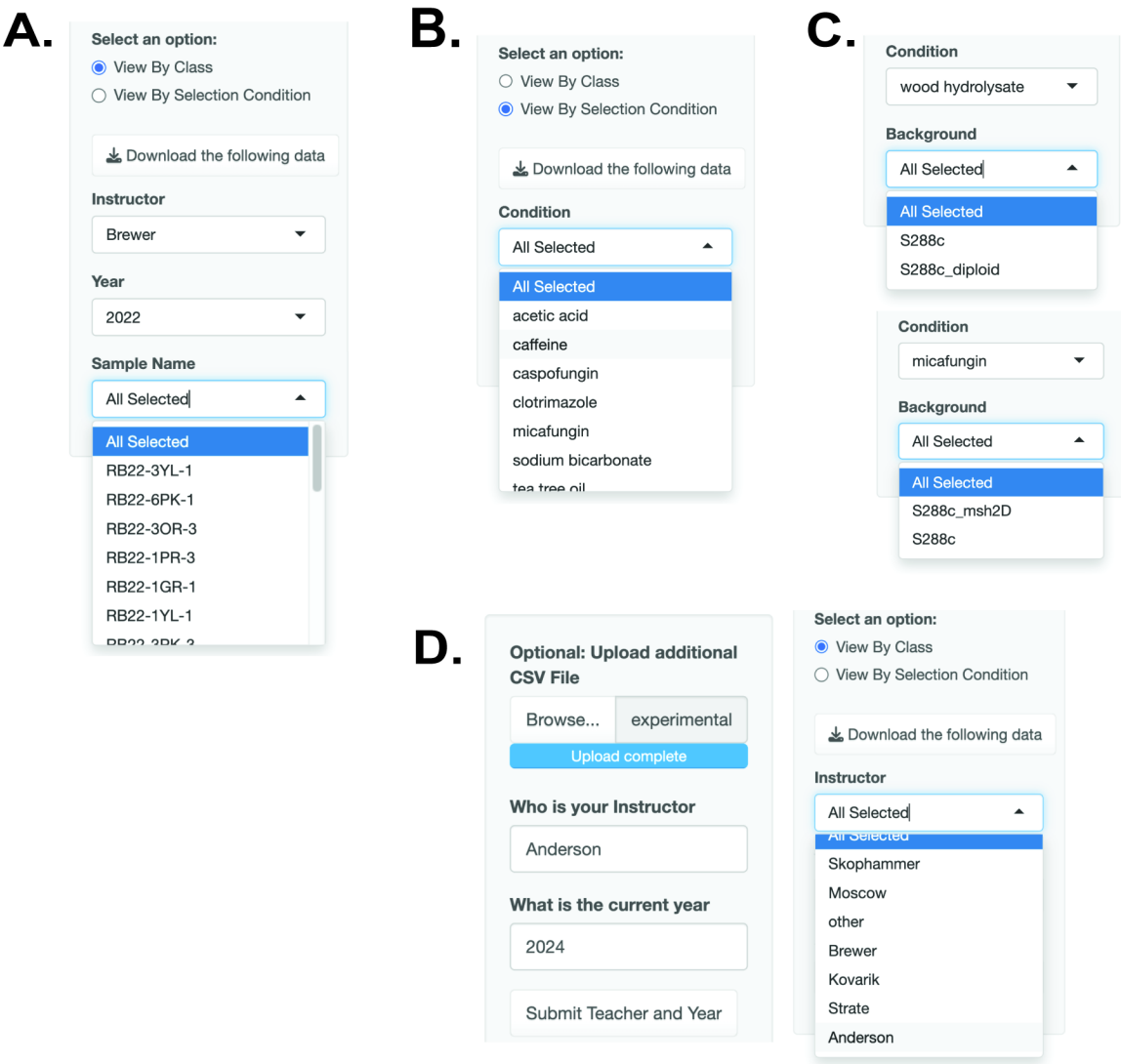
The yEvo Mutation browser contains two sections: the filtering column on the left and the visualizations on the right. Upon first loading, the app displays mutation data across the entire yEvo dataset. The user has the option to: 1.

change what data is being visualized (Filtering) and 2. change the visualization method. Both options are detailed below.

Filtering

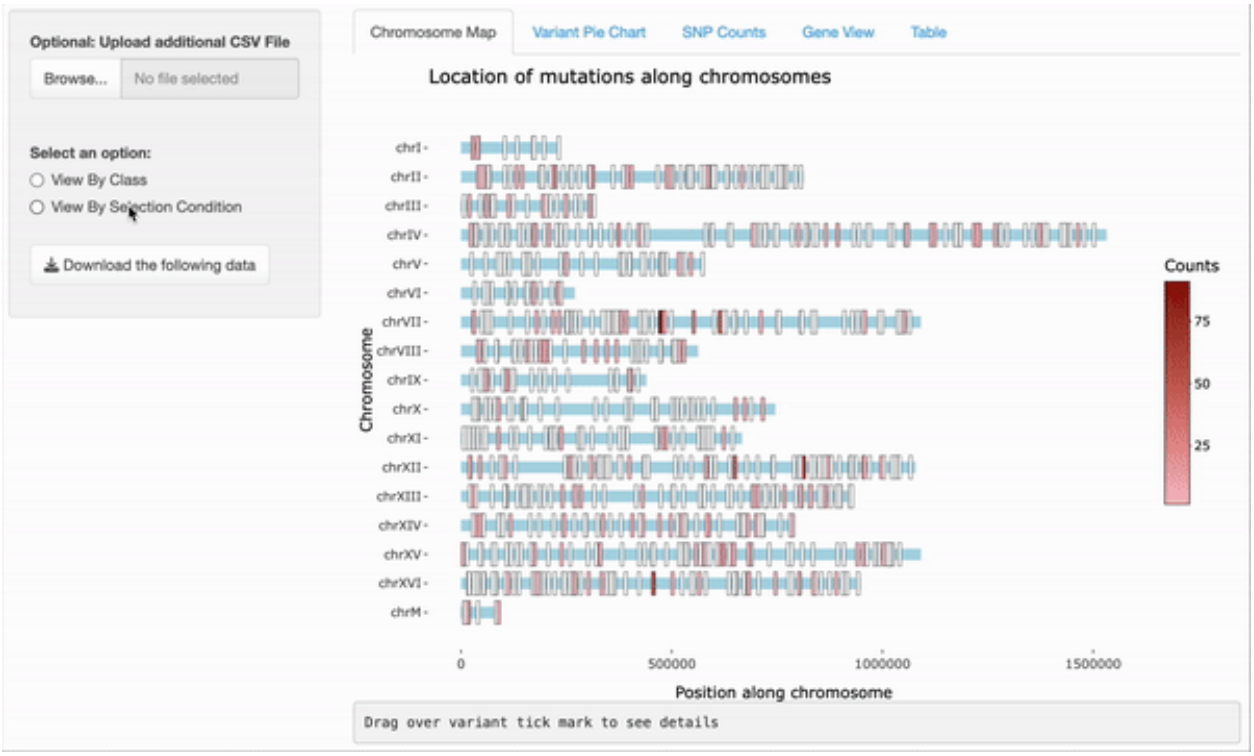
By default, the mutation browser displays all data. Users can filter this data in two ways: by student class or by selection condition.

As shown below, when selected, dropdowns appear corresponding to either option.




```
├── View by Class
│   ├── Instructor Dropdown
│   ├── Year Dropdown
│   └── Sample Name Dropdown
└── View by Condition
    └── Condition Dropdown
```

Below is an example of filtering on the Chromosome Map



Uploading Data

> [!IMPORTANT] Uploaded data is only available temporarily on the single instance of the app to which the data is uploaded. The data will **NOT** persist when the web app is closed and your data is not stored in the app.

By default, the mutation browser is preloaded with all mutation data collected from yEvo experiments since 2018. Users can explore this dataset, or upload their own temporarily.

To upload, users must have their mutation data (in .csv form) in the following format:

CHROM	POS	REF	ALT	INFO	ANNOTATION	REGION	GENE	PROTEIN	seq_file	background	condition	instructor	year	sample
chrII	627704	G	A	DP=120;VDI	missense	YBR202W	MCM7	A645T	2E4_S31_sa	S288c	clotrimazole	Skophammer	2018	RS18-AE-1
chrII	220031	A	T	DP=47;VDB	missense	YBL005W	PDR3	K854N	2E4_S31_sa	S288c	clotrimazole	Skophammer	2018	RS18-AE-1
chrXVI	50350	G	C	DP=53;VDB	missense	YPL260W	CUB1	E350Q	2E4_S31_sa	S288c	clotrimazole	Skophammer	2018	RS18-AE-1
chrIV	157691	C	G	DP=68;VDB	missense	YDL170W	UGA3	F458L	2E4_S31_sa	S288c	clotrimazole	Skophammer	2018	RS18-AE-1
chrIV	1474988	C	G	DP=90;VDB	missense	YDR516C	EMI2	A497P	2E4_S31_sa	S288c	clotrimazole	Skophammer	2018	RS18-AE-1
chrXI	619506	A	G	DP=110;VDI	intergenic	NA	NA	NA	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrVII	470845	G	C	DP=41;VDB	missense	YGL013C	PDR1	S485W	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrIV	419117	T	G	DP=48;VDB	missense	YDL019C	OSH2	N800H	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrVII	611240	C	A	DP=60;VDB	missense	YGR060W	ERG25	T226N	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrIV	892410	G	A	DP=64;VDB	missense	YDR213W	UPC2	G887D	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrIV	1461327	G	C	DP=68;VDB	missense	YDR506C	GMC1	N76K	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrXIV	405266	T	A	DP=70;VDB	missense	YNL118C	DCP2	K100M	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrXII	649144	C	G	DP=60;VDB	nonsense	YLR256W	HAP1	Y910*	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrXVI	743251	A	T	DP=49;VDB	synonymous	YPR108W	RPN7	L266L	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrXVI	798288	C	T	DP=87;VDB	synonymous	YPR134W	MSS18	L80L	A1Ryanfinal2	S288c	clotrimazole	Skophammer	2018	RS18-AF-1
chrXI	619506	A	G	DP=43;VDB	intergenic	NA	NA	NA	C11_2018_A	S288c	clotrimazole	Skophammer	2018	RS18-AF-2

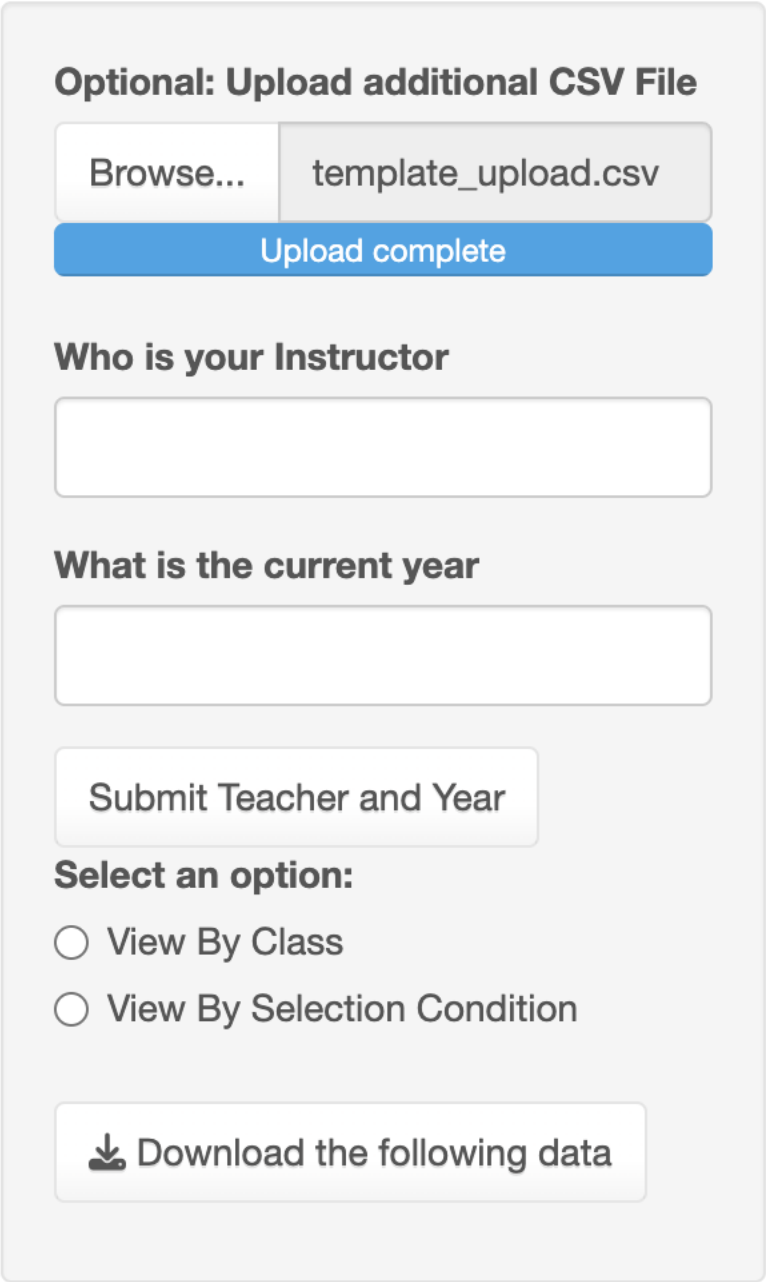
The CSV file **MUST** have the following columns:

- **CHROM**: the chromosome number in the form of `chr` followed by the chromosome number in uppercase

Roman numerals.

- **POS**: the position of the mutation on the chromosome.
- **REF**: nucleotide(s) of the reference genome
- **ALT**: nucleotide(s) of the mutant/evolved genome
- **ANNOTATION**: the type of variant. (Examples: missense, nonsense, synonymous, intergenic, rDNA, telomere, ncRNA)
- **REGION**: systematic name of the gene mutated. If the variant is not in a gene region, this should be “NA”
- **GENE**: common name of the gene mutated. If there is no common name for this gene, then add the systematic name here
- **PROTEIN**: result of the variant if it occurs in the coding region of a gene. (Example: ‘P327R’ for a missense or ‘P327P’ for synonymous). This should be “NA” if the variant is non-coding
- **background**: genetic background of the mutant (Example: S288C, S288C_diploid)
- **condition**: condition the mutant was evolved in
- **sample**: names of the mutated/evolved samples. the uploaded CSV can contain many individual sampeles

After uploading, the browser will ask for additional information (instructor and year):



Once you submit this information, the newly uploaded data will be available in the "View by Class" teacher dropdown menu.

> [!NOTE] To make your data persist in the mutation browser, you need to run the web app locally (see Running Locally below). Then, you can permanently edit the master file by appending your data so that your data will always appear in the browser.

Chromosome Map

The Chromosome Map is a linear representation of the 16 nuclear chromosomes along with the mitochondrial genome of *Saccharomyces cerevisiae*. This plot displays the locations of all mutated genes in the selected dataset, marked at their chromosomal positions. Tick marks along the linear chromosomes indicate these sites, and hovering

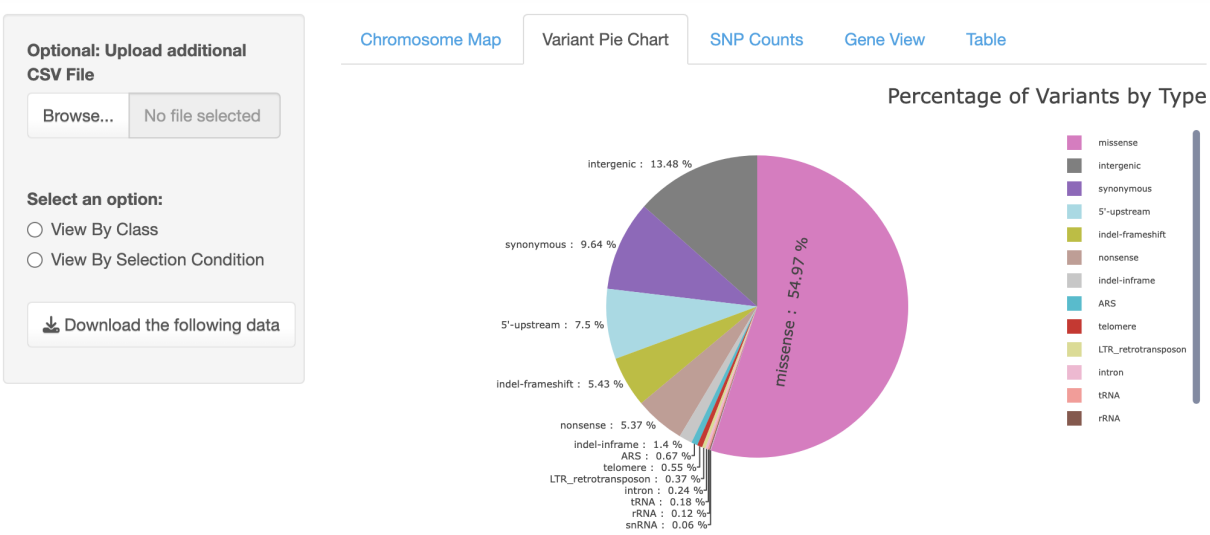
the mouse over each tick mark reveals the name of the mutated gene, as well as how many times it was mutated in repeat experiments.

Users can also zoom in on specific regions by clicking and dragging, offering a closer look at mutations in a given area.

Clicking on one of these tick marks that represents a gene will take you to Gene View (see below).

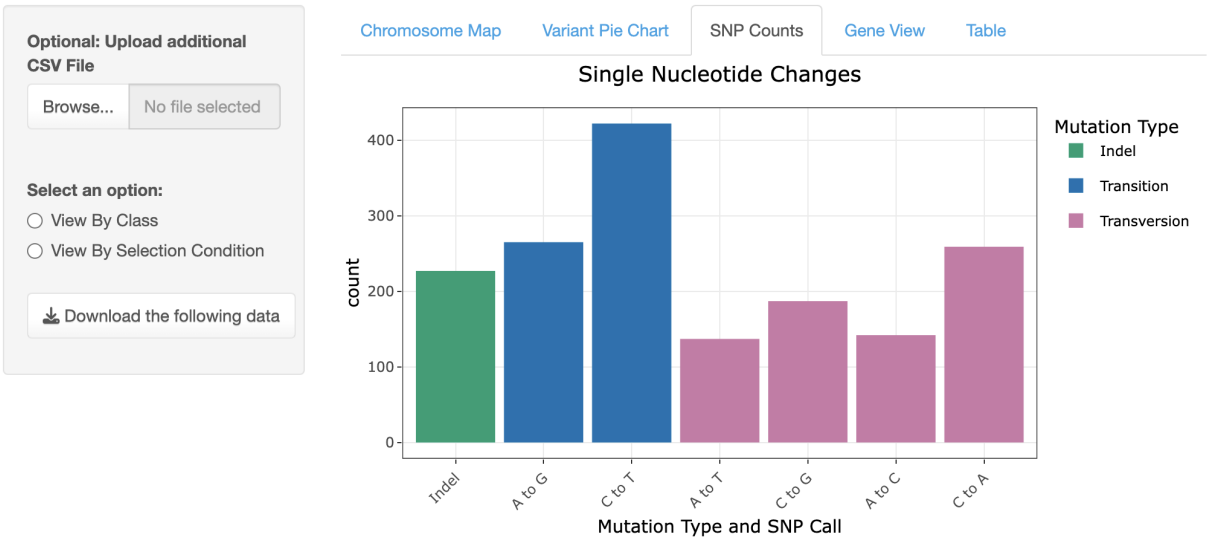
Variant Pie Chart

The Variant Pie Chart displays the distribution of mutation types identified by the sequencing analysis pipeline. Unlike the chromosome plot, which focuses solely on the locations of mutated genes, the pie chart encompasses all variants, including those in non-coding regions.



SNP Counts

This plot illustrates the distribution of single nucleotide polymorphisms (SNPs) and small insertions and deletions (indels). By categorizing the different types of SNPs, the mutation spectrum provides insights into the underlying biochemical processes that drive DNA mutations.



Gene View

Gene View is a linear lollipop plot that displays the position of coding mutations along the protein product of each mutated gene. In this view, users can select a mutated gene from a dropdown menu based on the selected dataset. Once a gene is selected, the plot reveals the length of the amino acid sequence and color-coded lollipops indicating where mutations occurred. The colors differentiate between mutation types, such as missense, nonsense, synonymous, or 5'-upstream mutations. The height of each lollipop reflects how frequently that particular site in the protein was mutated within the dataset.

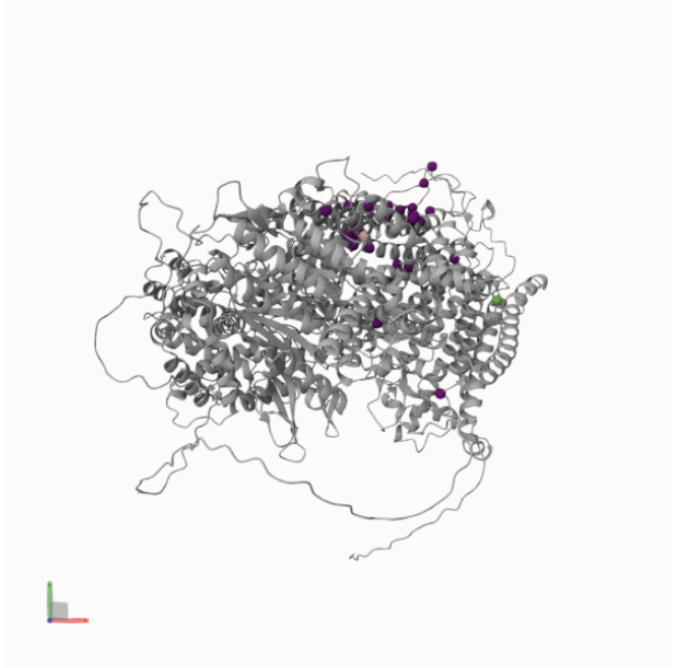
Protein View

Protein view shows the location and type of mutation on the 3D predicted protein stucture. Color coding matches the mutation types in Gene View. Users can select a mutation by clicking on that site in the linear track below the 3D rendering, or select all mutations in the filtered data by clicking the “Mutations” button. Protein domain information is also available, and can be highlighted on the structure by clicking on domains in the Pfam track. Confidence in the model prediction can be visualized using the PLDDT toggle switch.

Gene

FKS1

Take Screenshot



Position: 233 Amino Acid: ASP

Protein Info Card

Protein: FKS1

Description: Catalytic subunit of 1,3-beta-D-glucan synthase

Pathways:
N/A

PLDDT OFF

pLDDT Confidence Levels

Very High (pLDDT) > 90 | Confident (pLDDT) > 70 | Low (pLDDT) > 50 | Very Low (pLDDT) < 50

Running Locally

> [!IMPORTANT] R needs to be installed before you can run the yEvo Mutation Browser locally. All required R packages are listed in the "DESCRIPTION" file.

To make your data persist locally in the app, to contribute to the yEvo Mutation Browser, or to modify the mutation browser to better fit your needs or to another organism, you can run the yEvo Mutation Browser locally.

To do so, clone the repository:

```
git clone https://github.com/dunhamlab/yEvoMutBrowser.git
```

Then, navigate to the repository:

```
cd yEvoMutBrowser
```

To run the app, run the app using the following:

```
Rscript app.R
```

The app will tell you how to access it locally such as:

```
Listening on http://xxxxxxx
```

Copy and paste this link into a web browser of your choice to access the mutation browser.

Modifying to Other Organisms

The yEvo Mutation Browser can be modified for studying and visualizing mutations in other organisms.

The yEvo Mutation Browser is specific to *S. cerevisiae* because of the constants set in the `R/config.R` file:

- **`PATH_TO_VCF_CSV`**: the path to the CSV file containing the default VCF CSV dataset to load into the web app on startup.
- **`ORGANISM_GENE_INFO_PATH`**: the path to the CSV file with the information on all the genes in the organism being studied and visualized.
- **`ORGANISM_CHROMOSOME_INFO_PATH`**: the path to the CSV file with the information on the chromosomes in the organism being studied and visualized.
- **`ORGANISM_GENE_INFO_LINK`**: the link to a database with gene information. If this is set to `"NONE"`, then the "Learn about gene" button in the Gene View visualization will not appear.
- **`ORGANISM_GENE_INFO_LINK_FUNCTION`**: the function that generates specific links for each gene in an online gene information database for the "Learn about gene" button in the Gene View visualization. This is only used if `ORGANISM_GENE_INFO_LINK` is not `"NONE"`.

`ORGANISM_GENE_INFO_LINK_FUNCTION`

The `ORGANISM_GENE_INFO_LINK_FUNCTION` is a function that is passed two arguments by the Gene View visualization. The first is the genes info data table set by `ORGANISM_GENE_INFO_PATH` and the second is the gene selected by the user on the Gene View visualization gene selection dropdown. The function must return the link to the online gene database for the desired organism and the selected gene as a string. For an example, please see the default yEvo Mutation Browser link function for the *S. cerevisiae* gene database defined in `R/config.R`.